

ALEXANDER C. JENKINS

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ABOUT ME

I'm a theoretical physicist, working at the interface between *cosmology*, *astrophysics*, *high-energy physics*, and *quantum matter*. My research looks at new ways of probing the fundamental laws of Nature, whether that's using *gravitational waves* as powerful new astronomical messengers, or using cutting-edge *quantum technologies* to simulate the early Universe.

EMPLOYMENT

Gavin Boyle Fellow in Cosmology — *University of Cambridge* 2024–present

EPSRC Stephen Hawking Fellow 2025–present

Research fellow hosted in the [Kavli Institute for Cosmology Cambridge](#) (KICC) and the [Department of Applied Mathematics and Theoretical Physics](#) (DAMTP) | Fellow of [Selwyn College](#)

Postdoctoral Research Fellow — *University College London* 2021–2024

Led an international, interdisciplinary project to study vacuum decay with quantum analogue experiments and lattice simulations, as part of the [QSimFP Consortium](#) | Member of the [Cosmoparticle Initiative](#)

EDUCATION

PhD in Theoretical Physics — *King's College London* 2017–2021

Funded by competitive faculty scholarship | [Theoretical Particle Physics and Cosmology Group](#)

'*Cosmology and fundamental physics in the era of GW astronomy*', supervised by [Mairi Sakellariadou](#)

MSci in Astrophysics (Part III) — *University of Cambridge* 2016–2017

1st class | Ranked 5th in cohort | Elected a Bateman Scholar of [Trinity Hall](#) for 'excellent exam results'

Leonid Grishchuk Internship Programme — *Cardiff University* Summer 2016

Funded summer research internship in the [Gravitational Physics Group](#)

MA in Natural Sciences (Astrophysics) — *University of Cambridge* 2013–2016

1st class | Elected a Scholar of [Trinity Hall](#)

GRANTS AND FUNDING SECURED (POST-PHD)

- EPSRC Stephen Hawking Fellowship (PI, £506k fEC) 2025
3-year research council fellowship supporting 'visionary scientists working in theoretical physics'
Proposal ranked 2nd out of 26 by EPSRC prioritisation panel
- Gavin Boyle Fellowship, KICC and Selwyn College, Cambridge | 5-year institutional fellowship 2024
- UKRI Quantum Technologies for Fundamental Physics Grant extension (Co-I, £69k) 2023
Wrote successful proposal for 6-month funded extension to my QSimFP research

AFFILIATIONS

Scientific collaborations

* Quantum Simulators for Fundamental Physics (2021–present) * LISA Consortium (2018–present)

* Einstein Telescope (2021–present) * LIGO Scientific Collaboration (2016–2021)

Professional bodies

* Institute of Physics (MInstP; 2018–present) * Royal Astronomical Society (FRAS; 2020–present)

* Junior Member, European Astronomical Society (2020–present)

AWARDS AND ACHIEVEMENTS

- Winner, [Buchalter Cosmology Prize](#) (2nd Prize) | [UCL press release](#) 2023
International award recognising ‘ground-breaking theoretical, observational, or experimental work in cosmology that has the potential to produce a breakthrough advance in our understanding’
- Honorable Mention, [GWIC-Braccini Thesis Prize](#) | nominated for three further thesis prizes 2022
- Best Student Talk Prize at [BritGrav 21](#), sponsored by IoP Publishing 2021
Corresponding paper published in *Classical and Quantum Gravity* as an invited submission
- [King’s Education Award](#) for ‘extraordinary contributions’ to teaching 2020
Also nominated as a ‘Rising Star’ in 2019 (only PhD student nominee in my department)
- Bateman Scholar of [Trinity Hall](#) (Cambridge), recognising ‘excellent exam results’ 2017

RESPONSIBILITIES

Organisation of scientific meetings and seminars

- Lead organiser, [Kavli Focus Meeting on Gravitational Waves](#) 2025
Initiated and led the organisation of a local workshop bringing together an interdisciplinary mix of researchers across three different Cambridge departments, with funding secured from KICC
- Co-organiser, workshop on ‘Gravitational Waves and Dark Matter in Orbital Dynamics’ 2025
SOC member for workshop in Barcelona, funding secured from the [Gravity Theory Trust](#) (\$18k)
- Lead organiser, *UCL Cosmology/Extragalactic Seminars* 2022–2023
Developed an ambitious programme of in-person talks with speakers from across the UK and abroad, from what had previously been an online-only event due to COVID-19; secured and managed small grant (£1k) for speaker expenses and group lunches to encourage student participation
- Co-organiser, [London Cosmology Discussion Meetings \(LCDM\)](#) 2018–2021
Coordinated between five institutions to organise meetings at the Royal Astronomical Society on ‘Dark Matter in Cosmology’, ‘Neutrinos in Cosmology’, and ‘Cosmological Probes of New Physics’
- Organiser, *Theoretical Particle Physics and Cosmology (TPPC) Journal Club* 2021
- Organiser, *TPPC Gravity Meetings* 2018–2020
Initiated a regular series of meetings with internal and external speakers on gravitational physics

Refereeing and review work

- Expert referee for two ERC Consolidator Grants, one NSF Research Grant, 2019–present
one EPSRC Research Grant, one EPSRC Stephen Hawking Fellowship,
and one Royal Society Dorothy Hodgkin Fellowship
- Referee for 30 articles in Physical Review Letters, Nature Astronomy, 2020–present
Physical Review D, Journal of Cosmology and Astroparticle Physics,
European Physical Journal C, The Astronomical Journal, and Universe

Committees and institutional service

- Elector, Selwyn College Mastership Election 2024–2025
Contributed to interviews and selection process for the new head of my Cambridge college
- Selwyn College Governing Body and IT and Data Committee 2024–present
- Student representative, KCL Physics Department Research Committee 2019–2021

TEACHING AND SUPERVISION

University of Cambridge

2024–present

- Delivered three 1hr lectures for 3rd-year *Relativity* (Part II Physics/Astrophysics); student feedback highlighted my ‘very clear and engaging lecturing’
- Led small-group ‘supervision’ teaching in *Relativity* and *Statistical Physics* for 3rd-year Astrophysicists

University College London

2021–2024

- Lead supervisor of two MSc students: Phoebe Routh ([distinction](#), now a PhD student in Glasgow) and David Moody ([distinction](#) and awarded departmental prize, now working in data science)
- Postgraduate Teaching Assistant for 3rd-year *Physical Cosmology*: developed problem sets, delivered problem-solving tutorials, and contributed to marking of final exams

King’s College London

2017–2021

- Winner of a 2020 [King’s Education Award](#), recognising ‘extraordinary contributions’ to teaching
- ‘Rising Star’ nominee in the 2019 King’s Education Awards (only PhD student nominee in Physics)
- Co-wrote lecture notes for 3rd-year *General Relativity and Cosmology*
- Examples class demonstrator for numerous courses, including 4th-year *Astroparticle Cosmology*, 3rd-year *General Relativity and Cosmology*, 2nd-year *Astrophysics*, 1st-year *Mathematics for Physicists*, ...

SELECTED PUBLIC ENGAGEMENT

- Contributed to the art/science exhibition ‘*Cosmic Titans*’ at Lakeside Arts, Nottingham 2025
Interacted with artist Conrad Shawcross RA to influence his piece ‘*The Blind Proliferation*’; gave a [video interview](#) exploring the science behind Conrad’s work, featured in the exhibition and online
- My research and simulations featured in the documentary ‘*Do we live in a multiverse?*’ 2024
Aired on [French](#) and [German](#) TV and more than 5 million combined views on YouTube
- Led maths outreach activities at a local school as part of the [Cambridge NRIC project](#) 2024
- Invited speaker for the [Cambridge Astronomical Association](#) 2024
- YouTube video interview on ‘*Early Universe Cosmology in the Lab*’ 2023
- Outreach talk for alumni of UCL’s ‘[Introduction to Astronomy](#)’ course 2023
- Participated in five interviews for media pieces on my paper 2022
‘*Bridging the μHz gap in the gravitational-wave landscape with binary resonance*’
- Maths and physics tutor at Open Tutors London 2017–2020
Co-initiated tutoring programme for University of London students from under-represented groups
- Ran an interactive exhibit on Dark Matter at [Science Gallery London](#) 2019
- Local organiser, [Pint of Science Festival](#) 2018

SOFTWARE AND NUMERICS

- Author of Fortran lattice field theory code `lattice-fvd` and Python code [gw-resonance](#)
- Experience with advanced numerical methods including, e.g., Fourier and Chebyshev pseudospectral methods and symplectic integration
- Extensive experience in Unix environments (Ubuntu/MacOS), including in HPC settings; extensive Python experience (object-oriented programming; data handling and visualisation; [Jupyter](#), [NumPy](#), [SciPy](#), [h5py](#), [Astropy](#), [healpy](#), [sympy](#), ...); other languages and software include Fortran, C++, Mathematica, Git, MATLAB, SageMath, SQL, $\text{\LaTeX}$ (including [TikZ](#)), ...

SELECTED PUBLICATIONS

Citation statistics as of 19 August 2025 (data from [INSPIRE-HEP](#)):

Lead-author only	17 papers, 698 citations, h -index = 12
Short author list only	28 papers, 1,380 citations, h -index = 16
All publications	109 papers, 36,998 citations, h -index = 68

Lead-author papers (author list is ordered alphabetically in some cases)

- L1. **ACJ**, H. V. Peiris, and A. Pontzen, *Bubbles in a box: Eliminating edge nucleation in cold-atom simulators of vacuum decay* (2025), *Phys. Rev. A* **112**, 023318, [arXiv:2504.02829](#) [cond-mat.quant-gas]
- L2. **ACJ**, I. G. Moss, T. P. Billam, Z. Hadzibabic, H. V. Peiris, and A. Pontzen, *Generalized cold-atom analogues for vacuum decay* (2024), *Phys. Rev. A* **110**, L031301, [arXiv:2311.02156](#) [cond-mat.quant-gas] | [Letter](#)
- L3. **ACJ**, J. Braden, H. V. Peiris, A. Pontzen, M. C. Johnson, and S. Weinfurter, *Analog vacuum decay from vacuum initial conditions* (2024), *Phys. Rev. D* **109**, 023506, [arXiv:2307.02549](#) [cond-mat.quant-gas] | [Editor's Suggestion](#)
- L4. A. K.-W. Chung, **ACJ**, J. D. Romano, and M. Sakellariadou, *Targeted search for the kinematic dipole of the gravitational-wave background* (2022), *Phys. Rev. D* **106**, 082005, [arXiv:2208.01330](#) [gr-qc]
- L5. M. R. Mosbech, **ACJ**, S. Bose, C. Boehm, M. Sakellariadou, and Y. Y. Y. Wong, *Gravitational-wave event rates as a new probe for dark matter microphysics* (2023), *Phys. Rev. D* **108**, 043512, [arXiv:2207.14126](#) [astro-ph.CO]
Co-lead author with Markus Mosbech; I developed the core idea and led $\sim 50\%$ of the analysis
[Featured](#) in a [Royal Astronomical Society press release](#) at the 2023 National Astronomy Meeting
- L6. D. Blas and **ACJ**, *Bridging the μHz gap in the gravitational-wave landscape with binary resonance* (2022), *Phys. Rev. Lett.* **128**, 101103, [arXiv:2107.04601](#) [astro-ph.CO]
Awarded a [Buchalter Cosmology Prize](#) (2nd Prize), recognising 'potential for remarkable impact'
Altmetric [attention score of 395](#), in the top 0.3% of all publications ever tracked by Altmetric
[Featured](#) in the *Daily Express*, *Physics* magazine, *Big Think*, *SYFY wire*, and 40+ other outlets
- L7. D. Blas and **ACJ**, *Detecting stochastic gravitational waves with binary resonance* (2022), *Phys. Rev. D* **105**, 064021, [arXiv:2107.04063](#) [gr-qc]
- L8. **ACJ** and M. Sakellariadou, *Nonlinear gravitational-wave memory from cusps and kinks on cosmic strings* (2021), *Class. Quant. Grav.* **38**, 165004, [arXiv:2102.12487](#) [gr-qc]
[Invited submission](#) to CQG as winner of the Best Student Talk Prize at [BritGrav 21](#)
- L9. **ACJ** and M. Sakellariadou, *Primordial black holes from cusp collapse on cosmic strings* (2020), [arXiv:2006.16249](#) [astro-ph.CO]
- L10. **ACJ**, J. D. Romano, and M. Sakellariadou, *Estimating the angular power spectrum of the gravitational-wave background in the presence of shot noise* (2019), *Phys. Rev. D* **100**, 083501, [arXiv:1907.06642](#) [astro-ph.CO]
- L11. **ACJ** and M. Sakellariadou, *Shot noise in the astrophysical gravitational-wave background* (2019), *Phys. Rev. D* **100**, 063508, [arXiv:1902.07719](#) [astro-ph.CO]
- L12. **ACJ**, R. O'Shaughnessy, M. Sakellariadou, and D. Wysocki, *Anisotropies in the astrophysical gravitational-wave background: The impact of black hole distributions* (2019), *Phys. Rev. Lett.* **122**, 111101, [arXiv:1810.13435](#) [astro-ph.CO]

- L13. **ACJ**, A. G. A. Pithis, and M. Sakellariadou, *Can we detect quantum gravity with compact binary inspirals?* (2018), *Phys. Rev. D* **98**, 104032, arXiv:1809.06275 [gr-qc]
- L14. **ACJ**, M. Sakellariadou, T. Regimbau, and E. Slezak, *Anisotropies in the astrophysical gravitational-wave background: Predictions for the detection of compact binaries by LIGO and Virgo* (2018), *Phys. Rev. D* **98**, 063501, arXiv:1806.01718 [astro-ph.CO]
- L15. **ACJ** and M. Sakellariadou, *Anisotropies in the stochastic gravitational-wave background: Formalism and the cosmic string case* (2018), *Phys. Rev. D* **98**, 063509, arXiv:1802.06046 [astro-ph.CO]
Featured in PRD's 'kaleidoscope' for Sep 2018

Other selected papers (with summary of my main contributions)

- O1. J. W. Foster, D. Blas, A. Bourgoïn, A. Hees, M. Herrero-Valea, **ACJ**, and X. Xue, *Prospects for gravitational wave and ultra-light dark matter detection with binary resonances beyond the secular approximation* (2025), arXiv:2504.16988 [gr-qc] | Significant conceptual and methodological contributions throughout project
- O2. J. W. Foster, D. Blas, A. Bourgoïn, A. Hees, M. Herrero-Valea, **ACJ**, and X. Xue, *Discovering μ Hz gravitational waves and ultra-light dark matter with binary resonances* (2025), arXiv:2504.15334 [astro-ph.CO] | Significant conceptual and methodological contributions throughout project
- O3. A. Abac *et al.*, *The Science of the Einstein Telescope* (2025), arXiv:2503.12263 [gr-qc]
Coordinated and led writing of section on cosmic strings for large collaboration white paper
- O4. L. Zwick, D. Soyuer, D. J. D'Orazio, D. O'Neill, A. Derdzinski, P. Saha, D. Blas, **ACJ**, L. Z. Kelley, *Bridging the micro-Hz gravitational wave gap via Doppler tracking with the Uranus Orbiter and Probe Mission: Massive black hole binaries, early universe signals and ultra-light dark matter* (2024), arXiv:2406.02306 [astro-ph.HE] | Led analysis of stochastic gravitational-wave signals and implications for early-Universe physics, wrote corresponding sections of paper
- O5. N. Kouvatsos, **ACJ**, A. I. Renzini, J. D. Romano, M. Sakellariadou, *Unbiased estimation of gravitational-wave anisotropies from noisy data* (2023), arXiv:2312.09110 [astro-ph.CO] | Proposed new analysis method and led theoretical work; informal supervision of PhD student (N. Kouvatsos)
- O6. M. Branchesi *et al.*, *Science with the Einstein Telescope: a comparison of different designs* (2023), *JCAP* **07**, 068, arXiv:2303.15923 [gr-qc] | Contributed to stochastic background sensitivity analysis for different Einstein Telescope configurations, guiding further development of the proposal
- O7. S. Gasparrotto, R. Vicente, D. Blas, **ACJ**, and E. Barausse, *Can gravitational-wave memory help constrain binary black-hole parameters? A LISA case study* (2023), *Phys. Rev. D* **107**, 124033, arXiv:2301.13228 [gr-qc] | Significant conceptual and methodological contributions; informal supervision of PhD student (S. Gasparrotto)
- O8. P. Auclair *et al.* (LISA Cosmology Working Group), *Cosmology with the Laser Interferometer Space Antenna* (2022), *Living Rev. Rel.* **26**, 5, arXiv:2204.05434 [astro-ph.CO] | LISA white paper; contributed to section on cosmic strings, led analysis of related gravitational-wave anisotropies
- O9. A. I. Renzini, B. Goncharov, **ACJ**, and P. M. Meyers, *Stochastic Gravitational-Wave Backgrounds: Current Detection Efforts and Future Prospects* (2022), *Galaxies* **10**, 34, arXiv:2202.00178 [gr-qc]
Invited review article; major contributions to sections on gravitational-wave theory and sources
- O10. N. Bartolo *et al.* (LISA Cosmology Working Group), *Probing Anisotropies of the Stochastic Gravitational Wave Background with LISA* (2022), *JCAP* **11**, 009, arXiv:2201.08782 [astro-ph.CO]
LISA review paper; coordinator for 'topological defects' section, with further contributions to 'astrophysical sources' section

- O11. P. Auclair, J. J. Blanco-Pillado, D. G. Figueroa, **ACJ**, M. Lewicki, M. Sakellariadou, S. Sanidas, L. Sousa, D. A. Steer, J. M. Wachter, and S. Kuroyanagi (LISA Cosmology Working Group), *Probing the gravitational wave background from cosmic strings with LISA* (2019), **JCAP** **04**, 034, [arXiv:1909.00819 \[astro-ph.CO\]](#) | LISA review paper; significant writing contributions throughout
- O12. B. P. Abbott *et al.* (LIGO, Virgo), *Directional limits on persistent gravitational waves using data from Advanced LIGO's first two observing runs* (2019), **Phys. Rev. D** **100**, 062001, [arXiv:1903.08844 \[gr-qc\]](#) | Led interpretation of observational results in the context of cosmological and astrophysical source models, wrote corresponding section
- O13. B. P. Abbott *et al.* (LIGO, Virgo), *Search for the isotropic stochastic background using data from Advanced LIGO's second observing run* (2019), **Phys. Rev. D** **100**, 061101, [arXiv:1903.02886 \[gr-qc\]](#) Rapid communication | Led and wrote section on implications for cosmic string models

INVITED TALKS

Total of 46 invited talks across three continents

- **Numerical Simulations of Early Universe Sources of Gravitational Waves** *Aug 2025*
Nordita, Stockholm
- **Cosmology Seminar** *Jul 2025*
CERN, Geneva
- **Gravitational Wave Probes of Physics Beyond Standard Model** *Jun 2025*
University of Warsaw
- **Multiverse from Home** *Jun 2025*
Online
- **WE-Heraeus Seminar: New Windows on the Universe** *May 2025*
Kitzbühel, Austria
- **QSimFP Community Workshop** *Mar 2025*
University of Nottingham
- **Early Universe from Home** *Feb 2025*
Online
- **Theoretical Particle Physics Seminar** *Feb 2025*
University of Sussex
- **Joint seminar, LMU Cosmology and MPI Quantum Optics** *Jan 2025*
Ludwig Maximilian University of Munich
- **General Relativity Seminar** *Dec 2024*
DAMTP, University of Cambridge
- **Cosmology Journal Club** *Nov 2024*
University of Padova (online)
- **Particle Cosmology Seminar** *Nov 2024*
University of Nottingham
- **Cosmology/Extragalactic Seminar** *Oct 2024*
University College London
- **Cambridge-LMU Meeting** *Oct 2024*
Kavli Institute for Cosmology, University of Cambridge

- **GEMMA2 Workshop** *Sapienza University of Rome* Sep 2024
- **Majorana-Raychaudhuri Seminar** *Kolkata/Salerno (online)* Sep 2024
- **Cold atoms and molecules for fundamental physics** *Cambridge* Jul 2024
- **4th EuCAPT Symposium** *CERN, Geneva* May 2024
- **British Applied Mathematics Colloquium (BAMC)** *Newcastle University* Apr 2024
- **Cosmology Lunch Seminar** *DAMTP, University of Cambridge* Feb 2024
- **Gravitational-Wave Group Meeting** *Institute of Cosmology and Gravitation (ICG), University of Portsmouth* Jan 2024
- **Oberthaler Group Seminar** *Kirchhoff Institute for Physics (KIP), Heidelberg* Nov 2023
- **Cosmology from Home** *Online* Jul 2023
- **Quantum Simulators for Fundamental Physics Workshop** *Perimeter Institute for Theoretical Physics, Waterloo, Canada* Jun 2023
- **Astrophysics Seminar** *University of Leicester* May 2023
- **Cosmology Seminar** *Beecroft Institute, Oxford* May 2023
- **Theory Group Seminar** *Astroparticle and Cosmology Laboratory (APC), Paris* May 2023
- **Cosmology and Relativity Seminar** *Queen Mary University of London* Apr 2023
- **London Gravity Meeting** *Royal Society, London* Mar 2023
- **‘Dark Matters’ Workshop** *Université Libre de Bruxelles (ULB)* Nov 2022
- **London-Oldenburg Relativity Seminar** *University College London/University of Oldenburg (online)* Nov 2022
- **ICTP-AP Seminar** *International Centre for Theoretical Physics, Asia-Pacific (online)* Sep 2022
- **Quantum Simulators for Fundamental Physics Workshop** *Science Gallery London* Sep 2022
- **‘Gravitational-Wave Orchestra’ Workshop** *Université Catholique de Louvain, Belgium* Sep 2022
- **Theory Group Seminar** *Institute of High-Energy Physics (IFAE), Barcelona* May 2022

- **Quantum Technology Seminar** *May 2022*
London Centre for Nanotechnology, University College London
- **Cosmology/Extragalactic Seminar** *Nov 2021*
University College London
- **Theory Group Seminar** *Oct 2021*
Astroparticle and Cosmology Laboratory (APC), Paris
- **Ibarra Group Seminar** *Jul 2021*
Technical University of Munich (online)
- **Gravitational Wave Probes of Physics Beyond the Standard Model** *Jul 2021*
University of Warsaw (online)
- **London Cosmology Discussion Meeting (LCDM)** *Dec 2020*
Royal Astronomical Society, London (online)
- **Theoretical Cosmology Seminar** *May 2020*
Institute of Cosmology and Gravitation (ICG), University of Portsmouth (online)
- **Cosmology Seminar** *May 2020*
Beecroft Institute, University of Oxford (online)
- **London Cosmology Discussion Meeting (LCDM)** *Feb 2020*
Royal Astronomical Society, London
- **Gravitational Wave Probes of Fundamental Physics** *Nov 2019*
EuCAPT workshop, Amsterdam
- **Seminar** *Feb 2019*
Virtual Institute of Astroparticle Physics (online)
- **Cosmology Coffee Seminar** *Oct 2018*
Imperial College London